

1. Networking-Basics

What is Networking?

Networking connects multiple computers and devices to share data and resources.

Types

- LAN
- WAN
- MAN
- PAN

Benefits

- Resource Sharing
- Communication
- Internet Access

2. Osi-Model

Seven Layers

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

Purpose

Standard communication framework for computer networks.

3. TCP-IP

TCP/IP Model

Layers

- Network Access
- Internet
- Transport
- Application

Protocols

- TCP
- UDP
- HTTP
- HTTPS
- FTP
- SSH

4. Ip-Addressing

IPv4

Example

192.168.1.1

IPv6

Example

2001:db8::1

Types

- Public IP
- Private IP
- Static IP
- Dynamic IP

5. DNS-DHCP

DNS

Converts domain names into IP addresses.

Example

google.com

↓

142.250.xxx.xxx

DHCP

Automatically assigns IP addresses to devices in a network.